Relations and Functions

- **Exercise 1**
  Let
  $$A = \{1, 2, 3\}.$$
  Consider the relation
  $$R = \{(1,1), (2,2), (3,3), (1,2), (2,1)\}.$$
  Determine whether R is:
  1. reflexive ✓.
  2. symmetric ✓.
  3. antisymmetric ×.
  4. transitive ✓.
- **Exercise 2**
  Determine which of the following relations are equivalence relations.
  1. On $\mathbb{Z}$,
     $$x \sim y \iff x - y \text{ is even.} \quad \checkmark$$
  2. On $\mathbb{R}$,
     $$x \sim y \iff x \leqslant y. \quad \times$$
  3. On $\mathbb{R}$,
     $$x \sim y \iff |x| = |y|. \quad \checkmark$$
  4. On $\mathbb{Z}$,
     $$x \sim y \iff |x - y| \leqslant 1. \quad \times$$
- **Exercise 3**
  Decide whether each of the following relations is a partial order.
  1. On $\mathbb{R}$,
     $$x \sim y \iff x \leqslant y. \quad \checkmark$$
  2. On $\mathbb{R}$,
     $$x \sim y \iff x < y. \quad \times$$
  3. On the power set $\mathscr{P}(X)$,
     $$A \sim B \iff A \subseteq B. \quad \checkmark$$
  4. On the positive integers,
     $$a \sim b \iff a \mid b. \quad \checkmark$$
- **Exercise 4**
  Let
  $$A = \{1, 2, 3\}, \quad B = \{a, b, c\}, \quad X = A \cup B.$$

Consider the following relations on X. Each relation is specified by listing all ordered pairs (x, y) for which $x \sim y$.

Which of these relations define a function

$$f\colon X \to B.$$

1. $$R_1 = \{(1,a),(2,b),(3,c)\}. \quad \checkmark$$
2. $$R_2 = \{(1,a),(1,b),(2,c),(3,a)\}. \quad \times$$
3. $$R_3 = \{(1,a),(2,b)\}. \quad \times$$
4. $$R_4 = \{(1,a),(2,a),(3,a)\}. \quad \checkmark$$

- **Exercise 5**
  Let

$$f\colon \mathbb{R} \to \mathbb{R}, \quad f(x) = x^2.$$

Determine

1. The domain of f

$$\mathbb{R}.$$

2. The codomain of f

$$\mathbb{R}.$$

3. The image of f

$$[0, \infty).$$

Is the image of a function always equal to its codomain? No.

- **Exercise 6**
  Consider

$$f\colon \mathbb{R} \to \mathbb{R}, \quad f(x) = x^2,$$

and

$$g\colon \mathbb{R} \to [0, \infty), \quad g(x) = x^2.$$

Are f and g equal as functions? No.

Now consider

$$h\colon \mathbb{R} \to \mathbb{R}, \quad h(x) = |x|^2.$$

Are f and h equal as functions? Yes.